MINISTRY OF EDUCATION AND TRAINING
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MASTER OF SOFTWARE ENGINEERING (MSE)
CAPSTONE PROJECT PROPOSAL

Project Title:

IOT-BASED SMART CAMERA SYSTEM WITH FACE RECOGNITION AND ANOMALY DETECTION

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LIST OF ABBREVIATIONS

Acronym	Full form
IoT	Internet of Things
AI	Artificial Intelligence
YOLO	You Only Look Once (Object Detection Model)
MSE	Master of Software Engineering
CV	Computer Vision
CPU	Central Processing Unit
API	Application Programming Interface
REST	Representational State Transfer
SMART	Specific, Measurable, Achievable, Relevant, and Time-bound
RQ	Research Question
TP./Tp.	City

1 Introduction and Background

1.1 Problem Description

With the rapid development of smart home technologies, many households have adopted surveillance cameras to enhance security. However, most low-cost home camera systems only provide basic motion detection and video recording without intelligent identification or behavioral analysis capabilities. Current commercial systems such as Google Nest Cam and Ring Video Doorbell offer advanced features including face recognition and cloud-based alerts. Nevertheless, these systems are relatively expensive, closed-source, and rely heavily on cloud services, raising concerns regarding privacy, customization, and long-term costs. In many developing regions, families require a cost-effective and privacy-preserving solution that can:

- Identify registered household members
- Detect unknown individuals
- Recognize abnormal entry behaviors
- Provide real-time alerts

There remains a research and practical gap in designing a lightweight, edge-based IoT camera system capable of real-time identification and anomaly detection suitable for small households [1]. Addressing this issue is essential to enhance home security, reduce burglary risks, and promote affordable smart home adoption [2].

1.2 Research Objectives

The project aims to develop an IoT-based smart camera system with the following SMART objectives:

- Develop a real-time face recognition system achieving at least 90% identification accuracy under controlled indoor conditions.
- Implement an anomaly detection mechanism capable of detecting unknown individuals and entry at unusual hours [3].
- Ensure system response time is less than 5 seconds from detection to notification [4].
- Deliver a fully functional prototype within 14 weeks.

1.2.1 Literature Review and Gap Analysis

To address the research questions, a comparison of existing solutions and the proposed system is presented in Table 2.

Table 2: Comparison of Existing Solutions vs. Proposed Smart Camera

Feature	**Nest Cam**	**Ring Doorbell**	**OpenCV/Pi**	**Proposed System**
Face Recognition	Advanced	Cloud-based	Basic	Advanced (FaceNet)
Privacy	Cloud-only	Cloud-only	Local	Local (Edge AI)
Anomaly Detection	Limited	Limited	Manual	Automated (YOLOv8)
Cost	High	High	Low	Low-Moderate
Connectivity	Constant	Constant	Offline/Local	Hybrid

1.2.2 Research Questions

- *RQ1:* What existing face recognition and anomaly detection methods can solve this problem? What are their advantages and disadvantages?
- *RQ2:* What method and architecture will be proposed in this project? How will it be implemented on an IoT edge device?
- *RQ3:* How will the performance, accuracy, and usability of the system be evaluated?

1.3 Scope of the Project

Included:

- Face registration of household members and real-time face recognition.
- Unknown person detection and rule-based anomaly detection.
- Web dashboard for monitoring and notification system (email or app).

Excluded:

- Full smart home integration and large-scale cloud deployment.
- Emotion recognition and complex behavioral AI analysis.

2 Proposed Solution

2.1 Solution Description

The proposed system is an IoT-based smart camera solution integrating edge AI processing and backend services [1]. The system processes video streams locally (edge computing) to reduce latency and preserve privacy [5]. Recognition results are transmitted to the backend server for logging and notification, with an emphasis on a user-friendly interface for residents [6].

2.2 Software Architecture (General)

The system adopts a Client-Server architecture with Edge AI processing, as illustrated in Figure 1.

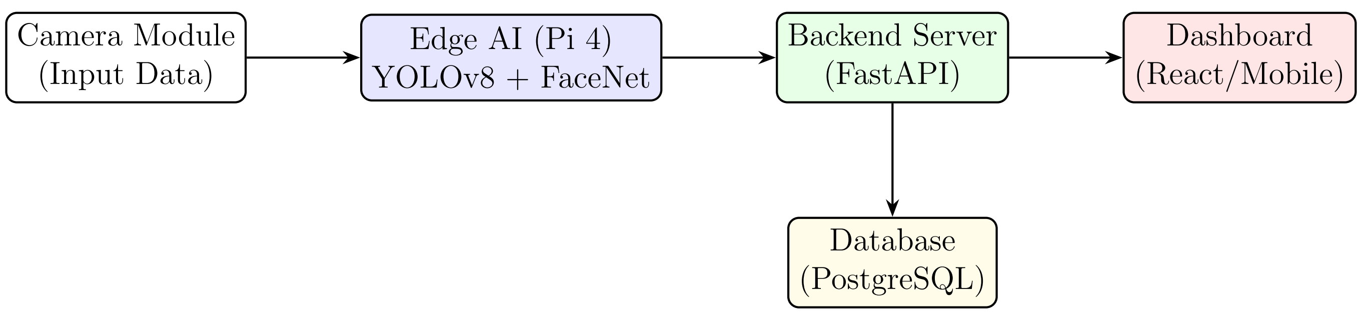

Figure 1: System Architecture Diagram

- **Edge Layer:** Face detection using YOLOv8 [7]; Face recognition using FaceNet embeddings.
- **Backend Layer:** REST API using FastAPI [8]; PostgreSQL database for logging.
- **Frontend Layer:** ReactJS/NextJS dashboard for monitoring alerts.

2.3 Technology and Tools Used

- **Hardware:** Raspberry Pi 4 Model B (8GB RAM), Raspberry Pi Camera Module V2 (8MP).
- **AI & Computer Vision:** Python 3.10, OpenCV 4.8, FaceNet (Keras-based), YOLOv8.
- **Backend:** FastAPI, PostgreSQL, SQLAlchemy.
- **Frontend:** Next.js, React Native (for mobile alerts).

3 Implementation Plan

3.1 Main Development Stages

1. Phase 1 – Requirements Analysis
2. Phase 2 – Literature Review and Model Comparison
3. Phase 3 – System Design
4. Phase 4 – Development and Integration
5. Phase 5 – Testing and Optimization
6. Phase 6 – Final Deployment and Documentation

3.2 Expected Schedule (Preliminary)

Table 3: Preliminary Project Schedule

Timeframes	**Activities**
Weeks 1–2	Requirement Analysis and Research
Weeks 3–4	Model Evaluation and Architecture Design
Weeks 5–8	Core System Development
Weeks 9–11	Integration and Dashboard Development
Weeks 12–13	Testing and Performance Evaluation
Week 14	Final Report and Presentation

3.3 Feasibility Assessment

- **Technical Challenges:** Low-light recognition accuracy, edge device computational limitations (CPU/RAM).
- **Mitigation:** Image preprocessing (normalization), lightweight models (YOLOv8-Nano), and multi-threading.
- **Resources:** Raspberry Pi 4, Sony IMX219 sensor, Labeled Faces in the Wild (LFW) dataset for pre-training, and locally captured dataset (family faces) for fine-tuning.

4 Expected Results

Upon completion, the project will deliver:

- A fully functional IoT smart camera prototype.
- Face recognition accuracy $\geq 85\%$.
- Alert response time ≤ 5 seconds.
- Web dashboard for monitoring and source code documentation.

5 Evaluation Plan

The system will be evaluated using:

- **Technical Metrics:** Accuracy, Precision, Recall, F1-score, Confusion Matrix [2].
- **Performance Metrics:** Detection Latency (ms), Identification Latency (ms), CPU/RAM usage [1].
- **Network Metrics:** Data throughput under Wi-Fi, 4G, and limited connectivity scenarios.
- **User Evaluation:** Usability testing (System Usability Scale) and user satisfaction survey targeting UI/UX efficiency [9].

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